

Pyrrmit – feasibility agent: benchmark report

Feasibility assistant for building projects in Sweden (planning law, PBL, Environmental Code, detailed development plans). Model: GPT-6 Luna via OpenRouter. Report generated on 2026-10-10.

Summary

- The system under test is a **tool-using agent** that explores a local corpus by itself: 15 Swedish statutes, the websites of all 290 municipalities (kommuner), 8,258 indexed detailed plans (detaljplaner). It answers with cited, clickable sources.
- Two benchmarks: **14 hand-written case studies** (real projects in the corpus municipalities) and **50 cases drawn from real rulings of the Land and Environment Court of Appeal (MÖD)**, out of 260 generated from 746 rulings.
- Each answer is scored on the verdict, the expected key points, the evidence actually consulted, how well its claims are grounded in the cited sources, latency and cost.
- Main result:** the reasoning effort of the exploration turns is the setting that matters. Raising exploration from *low* to *medium* makes the agent see much more of the expected evidence (0.53 → 0.70 on the MÖD cases), improves grounding (0.69 → 0.79) and key-point coverage (0.43 → 0.48), at the cost of ~17 s and 2.4× the price (≈ \$0.009 per question). A *high*-effort final answer brings nothing measurable.
- Weak spot:** on projects the court refused, the agent concludes “not feasible” in only 40–60 % of cases; it leans towards “feasible with conditions”. This is the first prompt fix to make.
- The MÖD cases are much harder than the 14 case studies, which saturate (verdict ≈ 1): they are the benchmark to use for the next decisions.
- Cost of the agent answers in this report: about \$4.14 (excluding the judge and case generation).

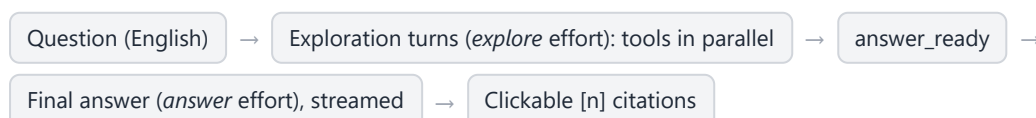
1. The system under test

1.1 Offline corpus

Source	Content	Processing
Statutes (Riksdagen open data)	15 consolidated texts: Planning and Building Act (PBL) and Ordinance (PBF), Environmental Code (MB), Land Code (JB), Real Property Formation Act, Joint Facilities Act, water and sewage (LAV, FMH), Cultural Environment Act...	One chunk per section (§), chapters and lettered sections handled
Websites of the 290 municipalities (Wikidata)	Pages on building permits, sewage, fees, plan requests, comprehensive plans, detailed plans + linked PDFs (plan descriptions, plan maps, fee schedules)	Crawl honouring robots.txt (1 request/s), PyMuPDF text, local OCR (EasyOCR) for scanned PDFs
Detailed plan registry	8,258 plans: status, date of legal force, plan map / plan description, extracted plan provisions (building area, ridge height, plot size...)	Rule-based, no LLM

In total 344,199 passages (chunks) in 31,497 documents, 275 municipalities represented. Retrieval: local embeddings `snowflake-arctic-embed-l-v2` (chosen on a retrieval benchmark: 0.92 recall@5) + BM25 on Swedish stems (SQLite FTS5), RRF fusion, municipality filter; all on disk (SQLite + vectors on the GPU, ~0.1 s per search).

1.2 Agent loop



Tool	Role
find_kommun	Place → municipality, and what the corpus holds for it
search	Hybrid search (semantic + Swedish BM25) over the statutes and/or a municipality
grep	Exact search: plan numbers, property designations, "nockhöjd", amounts
list_documents / read	List a municipality's documents, read a passage or a document
law_section	Exact text of a section ("PBL 9 kap. 4 §")
plans / plan_rules	Detailed plan registry and each plan's provisions as a table
view_pdf_page	Image of a PDF page for the model (plan map)

Every tool output carries passage identifiers ([c:id]); the answer must cite them for every rule, number or deadline. Two reasoning-effort settings: **explore** for the turns where the agent searches, **answer** for the final answer. They are the main subject of this benchmark.

2. How the benchmark was built

2.1 The 14 case studies

Written by hand from corpus documents and user feedback (questions such as "building permit procedure in Vallentuna", "maximum height in Veda", sewage, infrastructure). Three families: real projects on detailed plans (Kumla-Stensta, Mörby backe, Kristineberg), national + local rules (Attefall houses, shoreline protection, eco-village on farmland, private sewage, plot subdivision, demolition...), procedures (advance rulings, förhandsbesked). Each case has accepted verdicts, 3–5 key points, and **expected evidence** (a statute section or an exact excerpt of a municipal document) that the agent must have consulted.

2.2 Cases drawn from real rulings (MÖD)

1. **Collection:** 746 rulings of Mark- och miljööverdomstolen since 2011 (lagen.nu, whose robots.txt allows the ruling pages); 390 on our topics (building permits, detailed plans, shoreline protection, advance rulings, sewage...).
2. **Transformation** by the LLM: the facts become the question, asked as the applicant would *before* the decision (without the outcome or the reasoning); the actual outcome gives the accepted verdicts; the court's reasons give the key points; the statutes applied give the expected evidence.
3. **Checks:** purely procedural rulings dropped (standing, sanctions); questions that mention the court dropped; evidence = an exact phrase of at least 5 words of the **current** statute text (PBL chapter 9 was renumbered by SFS 2025:974 and some wording changed); rulings with no current provision kept without expected evidence (scored on verdict and key points).
4. **Sample:** 50 cases drawn at random (fixed seed), balanced between negative, positive and open outcomes, all with expected evidence.

Split of the 260 generated cases by accepted verdicts: not_feasible: 131, feasible / conditional: 70, conditional / unclear: 38, unclear: 8, feasible: 5. Generation cost: ~\$0.55.

2.3 Example cases

"I own one of the new plots in Kumla-Stensta, Vallentuna. Can I build a single-storey house with a 200 m² footprint, plus a 40 m² garage?"

Source: Detaljplan Kumla-Stensta, planbeskrivning (Vallentuna)

Accepted verdicts: not_feasible, unclear. **Expected evidence:** maximeras till 180; 9 kap. 3 §

Expected key points:

- The Kumla-Stensta detaljplan limits the building area (byggnadsarea) per property to 180 m² in the northern part and 160 m² in the southern part, so a 200 m² footprint exceeds it
- A separate ancillary building (komplementbyggnad) of up to 60 m² is allowed, so the 40 m² garage fits
- A new house requires a bygglov (PBL 9 kap. 3 §)
- There is no municipal water and sewage: a shared well / wastewater solution is needed

Agent, exploration low / answer medium: verdict "not_feasible" (correct), key points 0.25, evidence 0.50, 12 tool calls, 43 s. *Judge: The answer correctly identifies that the 200 m² house exceeds the plot limits, but misses the stated 60 m² allowance for an ancillary garage, the building-permit requirement, and the shared water and wastewater solution.*

Agent, exploration medium / answer medium: verdict "not_feasible" (correct), key points 0.75, evidence 0.50, 15 tool calls, 29 s. *Judge: The answer correctly identifies the house-area limit, garage allowance, and need for a house permit, but omits the shared water and wastewater requirement; its claim that the plankarta is legally binding is not supported by the cited excerpt.*

Agent, exploration medium / answer high: verdict "not_feasible" (correct), key points 0.62, evidence 0.50, 12 tool calls, 45 s. *Judge: The answer correctly identifies the house-area limit and garage allowance, but only implies rather than clearly states that a new house requires a permit and omits the shared water and wastewater requirement.*

"We want to build a holiday house about 40 metres from a lake shore in Ydre, outside any zoning plan. Is that possible?"

Source: Miljöbalken 7 kap.; Ydre page Strandskydd

Accepted verdicts: conditional, not_feasible, unclear. **Expected evidence:** 7 kap. 15 §; {"any": [{"law": "Miljöbalk", "section": "7 kap. 18 c §"}, {"kommun_code": "0512", "contains": "ansökan om strandskyddsdispens"}]}

Expected key points:

- Shoreline protection (strandskydd) covers 100 m from the shoreline (MB 7 kap. 14 §)
- New buildings are prohibited in the strandskydd area (MB 7 kap. 15 §)
- It is only possible with a strandskyddsdispens from the kommun, which requires special reasons (särskilda skäl, MB 7 kap. 18 c §)
- A bygglov is also needed (and a förhandsbesked can be requested outside detaljplan)

Agent, exploration low / answer medium: verdict "conditional" (correct), key points 1.00, evidence 1.00, 14 tool calls, 34 s. *Judge: The answer correctly explains that a house 40 metres from the shore would generally require a shoreline-protection exemption based on special reasons, as well as building approval, while adding three details not supported by the supplied excerpts.*

Agent, exploration medium / answer medium: verdict "conditional" (correct), key points 1.00, evidence 1.00, 30 tool calls, 37 s. *Judge: The answer correctly identifies the usual 100-metre protection, the building prohibition, the need for an exemption based on special reasons, and building-permit requirements; a few process or approval details go beyond the cited excerpts.*

Agent, exploration medium / answer high: verdict "conditional" (correct), key points 1.00, evidence 1.00, 33 tool calls, 47 s. *Judge: The answer captures the key conditions, but adds a few specific details not established by the reference or cited excerpts.*

"I want to convert two existing farm buildings on a rural agricultural property in the Boarp area of Båstad into homes. The buildings were built for agricultural use but have not been used for about 25 years, and the surrounding farmland is leased and farmed. The ground beneath the buildings is gravel and stone, and I understand it has not been classified as agricultural land in the relevant agricultural register."

Source: MÖD 2026:17 (P 13047-25, 2026-05-27): Förhandsbesked för ändrad användning av två ekonomibyggnader för jordbruk – actual outcome: MÖD set aside the municipality's positive preliminary decision and remitted the matter for the municipality to determine whether the land beneath the buildings is valuable agricultural land and, if so, apply the statutory protection test.

Accepted verdicts: unclear. **Expected evidence:** från allmän synpunkt tillfredsställande sätt genom att annan mark tas i anspråk

Expected key points:

- A positive förhandsbesked requires the proposed land use to be suitable under PBL 2 kap. 2 §.
- Under MB 3 kap. 4 §, valuable agricultural land may be used for development only to meet substantial societal interests where the need cannot be satisfactorily met by using other land.
- Existing agricultural buildings do not, by themselves, mean the land beneath them cannot be valuable agricultural land or that changing their use avoids the protection in MB 3 kap. 4 §.
- The municipality must establish whether the land is valuable agricultural land before deciding whether the statutory protection test applies.

Agent, exploration low / answer medium: verdict "unclear" (correct), key points 0.50, evidence 1.00, 14 tool calls, 30 s. *Judge: The answer recognizes unresolved planning and agricultural-land questions, but it does not clearly state the PBL suitability test or the full MB protection test, and it does not explicitly say the municipality must establish whether the land is valuable agricultural land.*

Agent, exploration medium / answer medium: verdict "conditional" (wrong), key points 0.62, evidence 1.00, 37 tool calls, 56 s. *Judge: The answer correctly explains the agricultural-land test but classifies the project too positively because the municipality has not established whether the land is valuable agricultural land or whether the proposal is suitable under PBL.*

Agent, exploration medium / answer high: verdict "conditional" (wrong), key points 0.50, evidence 1.00, 19 tool calls, 48 s. *Judge: The answer classifies the conversion as conditionally feasible, but the reference calls for an unclear outcome because the municipality must first establish whether the land is valuable agricultural land; the claims about durability and adaptation requirements are unsupported by the cited excerpts.*

"I own an attefall house and an older seasonal cottage in a coastal small-house area in Österåker; their lawful use has been residential accommodation complementary to a permanent home. I am considering renting them to tourists through online booking services for stays of a few days to a week, with a cleaning company cleaning after each stay; in 2024 they were rented about 34 and 20 times respectively, at a similar frequency for several years. Can I use the buildings this way?"

Source: MÖD 2026:13 (P 11254-25, 2026-05-21): Tillsyn med anledning av åtgärder på fastighet – actual outcome: MÖD dismissed the appeal, finding that the rentals required building permission for a change of use and that the municipality should continue its supervision rather than close the complaint without action.

Accepted verdicts: conditional, unclear. **Expected evidence:** för ett väsentligen annat ändamål än det som byggnaden

Expected key points:

- Under PBL 9 kap. 2 § första stycket 3 a, building permission is required when a building is taken into use for a substantially different purpose from its last lawful use.
- Whether the purpose is substantially different depends especially on the scale of the activity and its effects on the surroundings; use of the whole building may also matter.
- Residential use complementary to permanent living differs from temporary accommodation such as hotel or guesthouse use.
- Repeated paid short-term rentals arranged through booking services, together with cleaning after each stay, can indicate temporary accommodation when assessing the activity's character and scale.
- The ruling establishes that permission is required, not whether an application for permission would be granted.

Agent, exploration low / answer medium: verdict "unclear" (correct), key points 0.60, evidence 1.00, 9 tool calls, 28 s. *Judge: The answer captures the permit test and the relevance of repeated short lets and cleaning, but omits the effects-on-surroundings factor and does not clarify that the ruling establishes a permit requirement, not whether permission would be granted.*

Agent, exploration medium / answer medium: verdict "conditional" (correct), key points 0.80, evidence 1.00, 20 tool calls, 50 s. *Judge: The answer captures the conditional permit risk and the relevance of repeated short stays and cleaning, but omits effects on the surroundings and does not clearly state that the ruling decides only whether permission is required, not whether it would be granted.*

Agent, exploration medium / answer high: verdict "conditional" (correct), key points 0.80, evidence 1.00, 24 tool calls, 48 s. *Judge: The answer correctly identifies a possible change-of-use permit requirement and the relevance of repeated short stays and cleaning, but it does not clearly state that the ruling establishes a permit is required or address effects on the surroundings.*

3. How answers are scored

Measure	Definition	How
Verdict	The answer's verdict (feasible / with conditions / not feasible / unclear) is among the accepted verdicts	LLM judge
Key points	Each expected point scored 0 (missing or contradicted), 1 (partial), 2 (clear); scaled to 0–1	LLM judge
Evidence	Share of the expected evidence (statute section, document excerpt) that the agent <i>actually saw</i> in its tool outputs	Automatic, no LLM
Grounding	Share of the answer's factual claims supported by the text of the sources it cites	Second LLM judge, with the full source text
Unsupported claims	Rules or numbers found neither in the reference nor in the cited sources	LLM judge
Latency, 1st token, cost	Total time, time to the first word of the streamed answer, OpenRouter cost	Measured

Limits of the evaluation. The judge is the same model as the agent (GPT-6 Luna, medium effort): cheap but possibly lenient; its scores are for *comparing* settings, not absolute values. The key points of the MÖD cases are written by the LLM from the ruling: a point can be finer than what a good advisor would say. The judge gets the text of the cited sources so that a correct fact missing from the reference is not penalised. Each configuration was run 2 to 4 times: the pass-to-pass variance is given below. Cases run 3 in parallel: latencies include some contention.

4. Results

4.1 History on the 14 cases (5-municipality sub-corpus, successive versions)

Version	Key points	Evidence	Grounding	Latency	1st token	Cost
Classic RAG (1 search, high-effort answer)	0.56	0.58	0.84	30 s	–	\$0.0021
Agent v1 (high effort throughout)	0.75	0.86	0.68	64 s	–	\$0.0057
Agent v2 (citeable sources, medium effort)	0.77	0.96	0.86	39 s	39 s	\$0.0042
Agent v3 (low exploration, medium answer)	0.68	0.73	0.82	19 s	13 s	\$0.0025
Agent v4 (v3 + read the key municipal document)	0.65	0.84	0.84	23 s	17 s	\$0.0036

v1 → v2: citeable identifiers in every tool output and a “no claim without a source read” rule (grounding 0.68 → 0.86). v2 → v3/v4: parallel tools, `answer_ready`, streamed answer and *low* exploration effort: latency halved, but lower quality. Hence the matrix below.

4.2 Effort matrix (full corpus on disk)

14 case studies

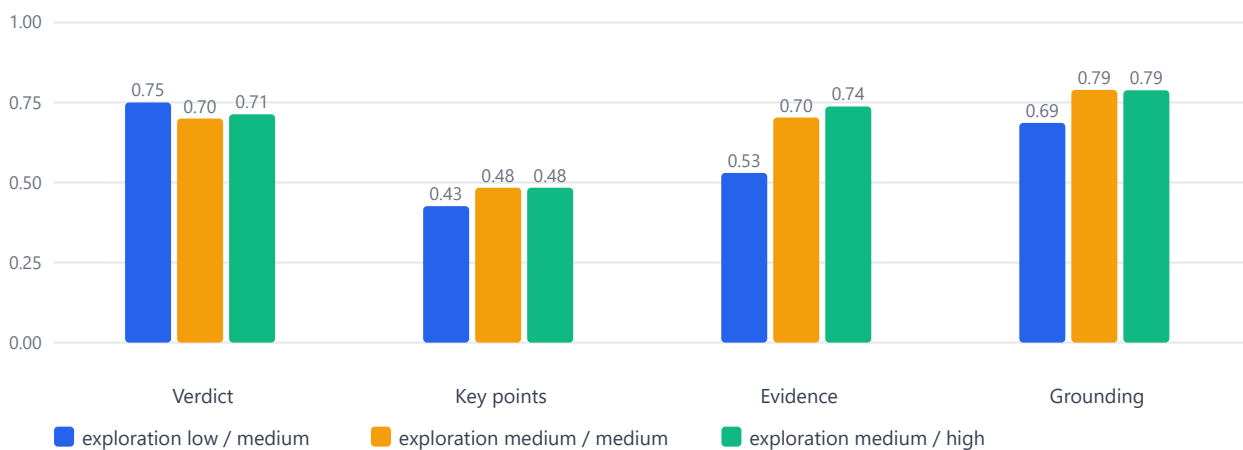
Configuration	n	Verdict	Key points	Evidence	Grounding	Tool calls	Latency	1st token	Cost / question
Agent, exploration low / answer medium	28	0.96	0.70	0.84	0.85	12.0	28 s	22 s	\$0.0036
Agent, exploration medium / answer medium	28	1.00	0.77	0.88	0.89	20.2	31 s	24 s	\$0.0071
Agent, exploration medium / answer high	56	0.98	0.76	0.87	0.85	20.2	38 s	31 s	\$0.0069

50 MÖD cases

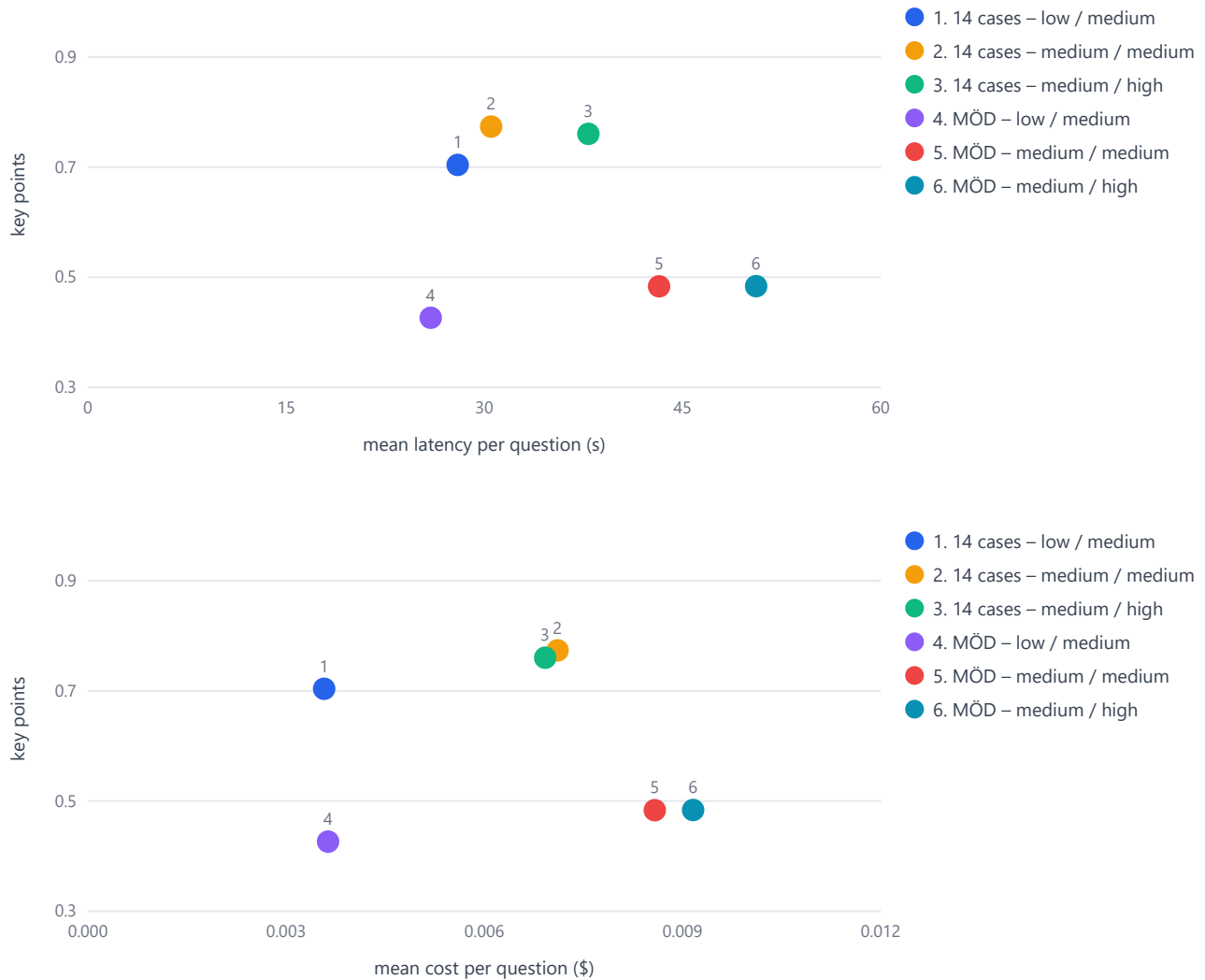
Configuration	n	Verdict	Key points	Evidence	Grounding	Tool calls	Latency	1st token	Cost / question
Agent, exploration low / answer medium	100	0.75	0.43	0.53	0.69	12.3	26 s	19 s	\$0.0036
Agent, exploration medium / answer medium	200	0.70	0.48	0.70	0.79	26.4	43 s	35 s	\$0.0086
Agent, exploration medium / answer high	150	0.71	0.48	0.74	0.79	26.7	51 s	42 s	\$0.0092

Best value of each column in green. n = number of answers (cases × passes).

4.3 Quality on the MÖD cases



4.4 Quality / speed / cost trade-off



Case by case on the 50 MÖD cases, **medium** exploration beats **low** on 25 cases, does worse on 3, and ties (± 0.05) on 22.

4.5 Verdict by actual outcome (MÖD cases)

Configuration	Project refused (correct verdict)	Project granted	Open outcome	On refusals, answers "with conditions"
Agent, exploration low / answer medium	0.59 (32)	0.74 (34)	0.91 (34)	0.28
Agent, exploration medium / answer medium	0.39 (64)	0.79 (68)	0.90 (68)	0.44
Agent, exploration medium / answer high	0.50 (48)	0.73 (51)	0.90 (51)	0.38

In brackets: number of answers. "Open outcome" = remitted or fact-dependent decision (accepted verdicts: with conditions / unclear).

4.6 Variance between passes

- Agent, exploration low / answer medium (14 cases): key points per pass 0.74 / 0.67

- Agent, exploration medium / answer medium (14 cases): key points per pass 0.75 / 0.79
- Agent, exploration medium / answer high (14 cases): key points per pass 0.77 / 0.78 / 0.71 / 0.78
- Agent, exploration low / answer medium (MÖD): key points per pass 0.43 / 0.43
- Agent, exploration medium / answer medium (MÖD): key points per pass 0.50 / 0.50 / 0.46 / 0.47
- Agent, exploration medium / answer high (MÖD): key points per pass 0.49 / 0.47 / 0.49

Typical gap between two identical passes: a few hundredths; a difference below ~0.05 between configurations is not significant.

Excerpt of an agent answer (mod-2026-17, Agent, exploration medium / answer medium)

Verdict: Feasible with conditions — converting the buildings may be possible, but a farm building's long period of non-use and the register status of its footprint do not by themselves establish that residential conversion is allowed.

Key rules and constraints

- **Expect a change-of-use building-permit review.** PBL requires a permit when a building is taken into use for a substantially different purpose from its last use or the purpose for which it was adapted under its latest permit. The separate exception for adding homes in an agricultural building applies only to the additional-dwelling trigger; it does not on its wording remove the change-of-purpose trigger. So the fact that these buildings were agricultural buildings, or that the proposed homes would be additional dwellings, is not by itself enough to make the conversion permit-free. [\[source\]](#) (PBL 9 kap. 15 §); [\[source\]](#) (PBL 9 kap. 17 §)
- **The planning position depends on the exact parcel.** Outside a detailed plan, a permit decision considers matters including area regulations, whether the proposal requires planning, and the PBL location and building requirements. Inside a detailed plan, the proposal and the existing property/building must meet the applicable plan conditions. [\[source\]](#) (PBL 9 kap. 57 §); [\[source\]](#) (PBL 9 kap. 56 §)
- **Boarp needs parcel-level checking.** Båstad's planning listing identifies a plan initiative for part of Boarp 2:32 and other properties at Norrviken, with work on consultation documents underway and a hotel, conference, spa, shop and craft-space purpose. The listing does not establish whether your buildings are inside that area, or what rules apply to your parcel. [\[source\]](#) (Båstad, "Del av Boarp 2:32 med flera f [...]")

5. Reading the results

- **Exploration effort matters, answer effort does not.** At *low*, the agent makes ~12 tool calls per question; at *medium*, ~20 (14 cases) to ~26 (MÖD), better targeted: it more often reads the decisive statute section and the relevant municipal document. Evidence seen: 0.84 → 0.88 on the 14 cases, 0.53 → 0.70 on the MÖD cases; grounding 0.69 → 0.79 on the MÖD cases; key points +0.05 to +0.07, beyond the pass-to-pass spread (~0.04). A *high* final answer changes neither key points nor evidence, and adds 7 s.
- **The price:** on the MÖD cases, latency 26 s → 43 s (first word of the answer: 19 s → 35 s), cost \$0.0036 → \$0.0086 per question. On the 14 cases, only +3 s. At 1,000 questions a month: ~\$9 instead of ~\$4.
- **Verdict calibration:** on the MÖD cases, the verdict is right for 90 % of open outcomes and ~75 % of granted projects, but only 40–60 % of refused projects: the agent answers "feasible with conditions" where the court said no (often: productive farmland, shoreline protection without a special reason, need for a detailed plan). More exploration reinforces this caution (more conditions found → "with conditions"), hence a slightly lower overall verdict score at *medium* (0.75 → 0.70) despite better evidence.
- **The MÖD cases are harder** than the 14 case studies: fine legal facts ("productive" land, "substantial" change of use, LIS conditions...) and key points written from the Court's full reasoning. The 14 cases saturate (verdict ≥ 0.96 everywhere) and no longer separate the settings.
- **Grounding:** 79–89 % of claims are supported by the cited sources; the rest is mostly general legal knowledge of the model, not re-read in the corpus.

6. Recommendations and next steps

- **Default app setting:** *medium* exploration, *medium* answer; keep *low* exploration as a “fast” mode (≈ 25 s). A *high* answer is not worth its cost.
- **Verdict prompt:** state explicitly that an unmet substantive condition (productive farmland without an essential public interest, shoreline protection without a special reason, a project that requires a detailed plan) makes the project “not feasible as described”, and do not turn every procedural requirement into “with conditions”. Measure the effect on the refused MÖD cases.
- **Benchmark:** use the 260 MÖD cases (samples of 50, 2 passes) for every prompt or model change; add “plan map only” cases now that the scanned PDFs are OCR'd, once map reading is in place.
- **Corpus:** the OCR of the 769 scanned PDFs is done (their text enters the index at the next rebuild); next, link each plan code to its area on the map (vision model) and each property to its applicable plan (municipal WMS / Lantmäteriet).
- **Judge:** have a domain expert review a sample of scores to calibrate the LLM judge.

Appendix: reproduce

```
python -m corpus.caselaw (MÖD rulings) · python -m agentic.cases_from_caselaw (cases) · python -m agentic.bench --systems agent --store disk --explore-effort medium --effort medium --cases-file cases_caselaw.jsonl --sample 50 --repeats 2 --workers 3 · python -m agentic.grounding <results> · python -m agentic.report --lang en. Result files used: bench-20261007-211220-agent-medium-explore-low.json, bench-20261007-214122-agent-medium-explore-low-cases_caselaw.json, bench-20261007-215229-agent-medium-explore-medium.json, bench-20261007-221849-agent-medium-explore-medium-cases_caselaw.json, bench-20261007-222545-agent-medium-explore-medium-cases_caselaw.json, bench-20261007-223027-agent-high-explore-medium.json, bench-20261007-225332-agent-high-explore-medium.json, bench-20261007-224026-agent-high-explore-medium-cases_caselaw.json, bench-20261007-230229-agent-high-explore-medium-cases_caselaw.json.
```